

Multiple Choice (10 marks)

1. A student made three successive weighings of an object as follows: 9.17 g, 9.15 g, and 9.20 g. What is the average weight of the object in milligrams?
 - (a) 9180 mg
 - (b) 9150 mg
 - (c) 9200 mg
 - (d) 9135 mg
 - (e) 9155 mg
2. Calculate the ratio of energy loss when a 4 MeV alpha particle travelling through carbon collides with nuclei (scattering) to the energy loss when it collides with bound electrons.
 - (a) 2.5×10^{-3}
 - (a) 5.5×10^{-3}
 - (a) 1.5×10^{-3}
 - (a) 1.0×10^{-3}
 - (a) 1.2×10^{-3}
3. The dissociation sequence of the polyprotic acid H_3PO_4 shows three Bronsted-Lowry acids. Rank them in order of decreasing strengths.
 - (a) H_2PO_4^- , HPO_4^{2-} , H_3PO_4
 - (b) HPO_4^{2-} , H_3PO_4 , H_2PO_4^-
 - (b) HPO_4^{2-} , H_2PO_4^- , H_3PO_4
 - (b) H_2PO_4^- , H_3PO_4 , PO_4^{3-}
 - (c) PO_4^{3-} , HPO_4^{2-} , H_2PO_4^-
4. What is the molarity of ethanol in 90-proof whiskey? Density of alcohol = 0.8 g/ml.
 - (a) 9.20 M
 - (b) 12.10 M
 - (c) 5.60 M
 - (d) 7.83 M
 - (e) 2.95 M

5. When air is let out of an automobile tire, the air feels cold to the touch. Explain why you feel this cooling sensation.
 - (a) The air feels cold because it mixes with colder atmospheric air
 - (b) The air feels cold due to a chemical reaction within the tire
 - (c) The air temperature falls because it is doing work and expending energy
 - (d) The cooling sensation is a result of the air's rapid expansion
 - (e) The cooling sensation is due to the air's humidity

True / False (10 marks)

Answer True or False

6. True or False: The spin angular momentum of a proton is $14.22 \times 10^{-35} \text{ J}\cdot\text{s}$, which is greater than the value for an electron.
6. True or False: The pH of a 0.25 M solution of Na_2CO_3 , a strong electrolyte, is approximately 11.83.
7. True or False: In a first-order reaction, the mass remaining after 58 s will be equal to half the initial mass of the starting material.
8. True or False: When 4.90 g of KClO_3 is heated, 20% of the original mass decomposes as evidenced by a weight loss of 0.384 g.
9. True or False: A 0.25 molal solution of NaCl can be achieved by adding 10 kg of water to 293 g of NaCl.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the mass of ice that can be melted at 0°C using the heat released from condensing 100 g of steam at 100°C .
11. Discuss the calculation of the mass of 1 liter of carbon monoxide (CO) at standard temperature and pressure (STP), considering its molecular weight and the ideal gas law.

End of Paper